WORKED SOLUTION

STEP I 1987, Question 14

Probability and Statistics • Geometric Series of Probabilities

> **STEP I 1987, Question 14**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/87-s1-q14

The key idea

Once the first game has been played, an undecided tournament has no freedom left at all. The winner of each game faces the referee, and if the tournament is to stay undecided the referee must win, so the next winner is forced. So everything is decided by who wins game 1 and by the single moment when a player first wins twice in a row. The probabilities then reduce to geometric series that cycle with period 3.

Solution

Throughout, in any game the winner of the previous game plays against its referee. The previous loser sits out as referee and *cannot win* the current game.

Part (i)

Game 1: A beats B, and B referees game 2. Game 2 is A against C, and C wins, which is possible. Game 3 is C against B, and B wins, which is also possible. But no one has won two games in a row, so after ACB the tournament is still undecided and must continue. Hence ACB is **not** the record of a complete tournament (it is only the start of one).

Part (ii)

After A wins game 1, game 2 is A against C with B refereeing. So B cannot win game 2, and ABB is **impossible**.

Part (iii)

As in (i), the first three games ACB are possible, and game 3 was C against B with A refereeing. So game 4 is B against A, and B can win it. Now B has won games 3 and 4 in a row, so the tournament ends. Hence $ACBB$ **is** a complete tournament, won by B in four games.

Undecided after 5 games

Suppose X wins game 1, where X is A or B, and let Y be the other one. If the tournament is to remain undecided, each game must be won by the player who was *not* the previous winner, and that can only be the previous referee. This forces the record

$$X\,C\,Y\,X\,C\,Y\,X\,\ldots$$

because each new winner is the previous game's referee. Concretely, game 2 is X v C (Y refereeing), so C must win. Game 3 is C v Y (X refereeing), so Y must win. Game 4 is Y v X, so X must win, and so on with period 3.

Game 1 can go either way. After that, each further game that keeps things undecided happens with probability $\frac{1}{2}$, since the games are independent and each player has an even chance. So

$$P(\text{undecided after } n \text{ games}) = \left(\tfrac{1}{2}\right)^{n-1}, \qquad P(\text{undecided after 5 games}) = \tfrac{1}{16}.$$

Who wins, and when

The tournament ends at game n (for $n \geq 2$) when it is undecided after $n-1$ games and the winner of game $n-1$ wins again. This has probability $\left(\frac{1}{2}\right)^{n-2} \cdot \frac{1}{2} = \left(\frac{1}{2}\right)^{n-1}$. The champion is the winner of game $n-1$, that is, the $(n-1)$th letter of the forced sequence. Write $k = n-1$. Then:

k mod 3	1	2	0
champion if A won game 1	A	C	B
champion if B won game 1	B	C	A

So the tournament ends at game $k+1$ with probability $\left(\frac{1}{2}\right)^{k}$. If $k \equiv 2 \pmod 3$ this whole probability belongs to C. Otherwise it is split equally between A and B, giving $\left(\frac{1}{2}\right)^{k+1}$ each.

Ask yourself Why does C get the whole probability at $k \equiv 2$, but only half at other times?

C is always the second letter of the forced sequence, whoever won game 1. So both branches give the win to C at the same moments. A and B swap roles between the two branches, so each of them collects only the branch in which they occupy that position.

Winning in 5 or fewer games

Here $n = 2, 3, 4, 5$, so $k = 1, 2, 3, 4$:

$$P(A \text{ wins in} \le 5) = \underbrace{\tfrac{1}{4}}_{k=1} + \underbrace{\tfrac{1}{16}}_{k=3} + \underbrace{\tfrac{1}{32}}_{k=4} = \tfrac{11}{32},$$
$$P(B \text{ wins in} \le 5) = \tfrac{11}{32} \quad \text{(by the symmetry between } A \text{ and } B\text{)},$$
$$P(C \text{ wins in} \le 5) = \underbrace{\tfrac{1}{4}}_{k=2}.$$

As a check, $\frac{11}{32} + \frac{11}{32} + \frac{8}{32} = \frac{15}{16} = 1 - \frac{1}{16}$, which agrees with the probability of being undecided after 5 games.

Eventual winner

Because $P(\text{undecided after } n \text{ games}) = \left(\frac{1}{2}\right)^{n-1} \to 0$, the tournament ends with probability 1. So we just add over all $k \ge 1$. Each residue class gives a geometric series with ratio $\frac{1}{8}$:

$$P(C \text{ wins}) = \tfrac{1}{4} + \tfrac{1}{32} + \tfrac{1}{256} + \cdots = \frac{1/4}{1 - 1/8} = \frac{2}{7},$$
$$P(A \text{ wins}) = \underbrace{\left(\tfrac{1}{4} + \tfrac{1}{32} + \cdots\right)}_{k\equiv 1} + \underbrace{\left(\tfrac{1}{16} + \tfrac{1}{128} + \cdots\right)}_{k\equiv 0} = \frac{1/4}{7/8} + \frac{1/16}{7/8} = \frac{2}{7} + \frac{1}{14} = \frac{5}{14},$$

as required. By symmetry $P(B \text{ wins}) = \frac{5}{14}$ as well, and indeed $\frac{5}{14} + \frac{5}{14} + \frac{2}{7} = 1$.

So C, who sits out the first game, is at a disadvantage: $\frac{2}{7} \approx 0.286$ against $\frac{5}{14} \approx 0.357$ for each of the others.

Checks

- The three probabilities of winning within 5 games add to $\frac{15}{16}$, and the eventual ones add to 1.
- An exact enumeration over all possible game sequences, using fractions, confirms $\frac{1}{16}$, $\frac{11}{32}$, $\frac{11}{32}$, $\frac{1}{4}$. Run to 60 games, it gives $0.357142\ldots$ for A and B and $0.285714\ldots$ for C.
- A quick alternative for C: let c be the probability that C wins. After game 1, C wins at once with probability $\frac{1}{4}$. If the play gets back to the same situation three games later (probability $\frac{1}{8}$), we start again, so $c = \frac{1}{4} + \frac{1}{8}c$, giving $c = \frac{2}{7}$.

Common slips

- Saying ACB is a valid record: every game in it is possible, but nobody has won twice running, so the tournament isn't over.

- Forgetting that the previous loser referees, and so can't win the next game (this rules out ABB).
- Giving C only half the probability at $k \equiv 2$: C wins at those moments in *both* branches.

Beyond STEP

The tournament is a Markov chain: the state after each game is just "who won last, and who is refereeing", and the champion is decided when the chain is absorbed. The first-step argument in the Checks section, writing an equation for a probability by conditioning on what happens next, is the standard way to find absorption probabilities, and it avoids summing series. Grimmett and Welsh, *Probability: An Introduction*, introduces Markov chains and absorption at first-year level.

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